Sahaj Bhadja

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Summary

Software engineer with 2+ years designing, developing, and testing dependable systems using **C++, Python, and Java**; experienced in system-level simulation, requirements analysis, algorithm development, verification, debugging, and Agile collaboration.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed and developed reliable Python software from stakeholder requirements through implementation and rigorous testing, processing 5,000+ recurring workflows with reconciliation, error handling, and 99%+ production reliability.
- Improved system performance for 500+ users by carefully analyzing software behavior, optimizing algorithms and interfaces, and reducing latency from 1.2 seconds to 350 milliseconds through measured verification.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed and tested Python services handling 10,000+ daily requests, implementing concurrent processing, failure isolation, and validation while maintaining quality and timely delivery for 150+ internal users.
- Collaborated with engineers and business stakeholders to clarify formal software requirements, coordinate feature development, resolve defects, and document releases, reducing feedback turnaround 60% in an Agile environment.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built extensive autonomous-driving software pipelines processing 100,000+ trajectory logs, improving long-horizon planning 18% through Python-based model development, algorithm evaluation, and rigorous, reproducible closed-loop system-level simulation testing.
- Designed a spatial-temporal encoder integrating road geometry, lane direction, and vehicle trajectories, then verified closed-loop behavior across 10,000 controlled simulation scenarios, improving overall planning stability 15%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered cross-platform software with object-oriented design patterns and 20+ automated unit and integration tests across Windows, Linux, and macOS, reducing technical debt 20% and response times 30%.
- Coordinated requirements, formal design reviews, defect resolution, and Agile delivery with Microsoft and GitHub engineers, documenting test results and improving distributed data-processing efficiency 68% across international teams.

Pandit Deendayal Energy University **India**
Research Fellowship *January 2023 – May 2024*

- Analyzed existing distributed consensus mechanisms and developed an algorithm reducing centralized-control and quantum-risk exposure 25% in rigorous benchmark simulations, applying systematic research, software design, and quantitative evaluation.
- Integrated five lattice-based cryptographic primitives into a secure system design, co-authoring a peer-reviewed technical paper and presenting detailed implementation findings at IISc Bengaluru through an IEEE publication.

Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript, SQL, Shell
Software Engineering: System requirements, software requirements, object-oriented design, algorithms, data structures, design patterns, interfaces
Verification: Unit testing, integration testing, system testing, simulation, validation, debugging, root-cause analysis, regression testing
Tools: Git, Linux, Unix, Jenkins, PyTest, Mockito, Docker, Jira, CI/CD, version control, build automation
Collaboration: Agile, Scrum, design reviews, technical documentation, customer communication, coordination, leadership, proactive problem-solving

Projects

CPU Scheduling Simulator Live | *C++, Algorithms, System Testing*

- Implemented six C++ scheduling algorithms with edge-case handling, interactive controls, and Gantt visualization, verifying correctness across 1,000+ process combinations while reducing average turnaround time 25% and debugging 20%.

Linguistix | *Python, AWS, Automated Verification*

- Developed a containerized Python data-processing service with automated build, validation, and deployment checks, achieving 87% measured accuracy and dependable production operation through repeatable CI/CD-based verification on AWS.

CaptionCraft Live | *Python, PyTorch, Performance Evaluation*

- Designed a repeatable evaluation framework across four software models, 10,000 images, and 100,000 captions, analyzing test results and failure patterns to improve measured system accuracy from 81% to 96%.